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A. Appendix A: prompt details, generation parameters and repetition penalty

This appendix reports the prompt templates used in our experiments, together with the generation hyperparameters adopted for each model and condition.

A.1. Prompt Templates

Two prompt variants were used. The **baseline** prompt asks the model to describe the target word freely, with no mention of forbidden words. The **prompted** condition additionally instructs the model to avoid the forbidden words and their morphological relatives.

User Message — Baseline
Descrivi la seguente parola in modo che qualcuno possa indovinarla. Parola da descrivere: [target word] Scrivi in output solamente la descrizione senza nessun testo di contorno. Considera che hai un limite di 30 parole, quindi cerca di dare una descrizione quanto più breve possibile. Descrizione:
<i>Describe the following word so that someone can guess it.</i> <i>Word to describe: [target word]</i> <i>Output only the description with no surrounding text. You have a limit of 30 words, so try to give as brief a description as possible.</i> <i>Description:</i>
User Message — Prompted condition
Descrivi la seguente parola in modo che qualcuno possa indovinarla, ma NON usare nessuna di queste parole o nessuna parola con la stessa radice: [fw ₁ , fw ₂ , . . .]. Parola da descrivere: [target word] Scrivi in output solamente la descrizione senza nessun testo di contorno. Considera che hai un limite di 30 parole, quindi cerca di dare una descrizione quanto più breve possibile. Descrizione:
<i>Describe the following word so that someone can guess it, but do NOT use any of these words or any word with the same root: [fw₁, fw₂, . . .].</i> <i>Word to describe: [target word]</i> <i>Output only the description with no surrounding text. You have a limit of 30 words, so try to give as brief a description as possible.</i> <i>Description:</i>

Table 2

Prompt templates for the **baseline** (top) and **prompted** (bottom) conditions. English translation in italics.

A.2. Generation parameters and repetition penalty

All models were run in greedy decoding mode (`do_sample=False`). Table 3 summarises the generation parameters used per model and condition.

Model	max_new_tokens	rep. penalty
<code>gemma-3-4b-it</code>	80	1.0
<code>gpt-oss-20b (reasoning)</code>	2048	1.2
<code>gpt-oss-20b (no reasoning)</code>	80	1.0

Table 3

Generation hyperparameters per model and condition.

For `openai/gpt-oss-20b` in the reasoning condition, the assistant turn was left open so the model could produce a chain-of-thought trace before committing to a final answer. In the no-reasoning condition, the assistant turn was instead prefilled with a channel token that routes the model directly to its final-answer channel, bypassing the reasoning trace. A repetition penalty of 1.2 was applied exclusively in the reasoning condition to mitigate degenerate repetition loops that can arise during long generations; no penalty was applied otherwise.

B. Appendix B: a priori constrained generation vs online

Both the *a priori* and the inference-time (online) approaches enforce the taboo constraint by operating on stems and lemmas of the forbidden words, using the same morphological pipeline (`spaCy it_core_news_sm + nltk Italian Stemmer`). They differ fundamentally in *when* and *how broadly* the constraint is applied.

A priori censorship Before generation begins, the method computes a static set of banned token IDs: for each forbidden word, all stems and lemmas are extracted, and every vocabulary token whose surface form shares a prefix with any of them (in either direction) is masked by forcing its logit to $-\infty$ for the entire generation. This static mask is context-free: a token is banned regardless of whether its occurrence in the output would actually constitute a violation.

The consequences can be severe. Consider the target word *accipicchia* (an Italian exclamation of surprise), whose forbidden list includes *per la miseria* (a common interjection meaning "what a misery"). Because the phrase contains the function words *per* ("for") and *la* ("the"), the stemmer extracts them as forbidden stems, causing 2171 vocabulary tokens to be masked — including common prepositions, determiners, and unrelated content words that merely share a surface prefix with *per* or *la* (e.g. tokens such as `_permitting`, `_lancé`, `larghezza`). Across the full dataset of 194 items, the *a priori* approach bans a cumulative total of 90 957 tokens (on average 469 per item, out of a vocabulary of 262 145).

Inference-time constrained generation The online approach bans *no tokens* before generation begins. Instead, it generates freely token by token, accumulates tokens into complete words, and only when a word boundary is detected checks whether the completed word matches